Science Olympiad — BullSO Div C 2026

Exam Preparation

You will need:

1. Folders for each of the teams to hold the tests
2. Sufficient copies of the test for all teams. They don't need to be stapled.
3. Multiple timers which have a lap function on them - ideally one per volunteer. The timer app on an iPhone or Android Phone that has a stopwatch function with lap function is sufficient.

Before the event begins:

1. Practice starting the timers and using the lap function to record the times. Make sure volunteers understand how to use the lap function and are not accidentally stopping the timer completely.
2. Memorize the answer to the timed question.
3. Check to make sure that this key matches the test you are proctoring.
4. Place one copy of the test for each team in the provided folders with the first page outside the folder.
5. Adjust desks and chairs – teams may have up to 3 students for this event.

Running the Event

1. When the students enter the room, instruct them to sit down, DO NOT OPEN THE FOLDER, and put their names, school name and school number on the first page.
2. Encourage them to write their team number on all the other pages AFTER they begin the test. This way if their papers gets separated from each other we can make sure to give them credit.
3. **CRITICAL:** Check to see that students have ONLY brought
    i. Something to write with (pencils, pens, erasers, highlighters)
    ii. Non-programmable, non-graphing calculators.
    iii. If the student has a smart watch (Apple watch, Samsung Gear, etc.) they will need to put it away.
4. Instruct the students that if they answer the timed question within 10 minutes, they can be awarded a bonus if they solve the timed question with no more than 2 letters incorrect.
    i. When they have a solution for the cryptogram they should raise their hand.
    ii. Let them know that you will announce when the 10-minute time is up. After the first 10 minutes, no additional bonus points will be awarded.
    iii. When you see a team raise their hand, hit the LAP function and head to the team.
    iv. Determine if their answer is correct (see next page for grading), If so, write the time on their score sheet.
    v. If their score is incorrect (more than 2 letters incorrect), tell the team that the answer is wrong, but DO NOT tell them what is wrong. They can continue to work on the question and raise their hand again to be checked. A team has an unlimited number of attempts during the 10-minute bonus.
5. Tell the teams that they do not have to fill in the frequency table. It is simply there as an aid to them solving the cryptogram. It will not be graded.
6. When the timers hit the 10-minute point, announce that no bonus points will be awarded and put away the timers. The students may continue to work on the question, but they may not receive any extra points.
7. A team is not restricted to only the timed question during the 10 minutes. They can move on or split up the work if they would like, but it is in their best interest to try for the bonus.
8. When time is up, have the students put writing instruments down and put their answer pages back into the folder in the correct order.

How to grade

1. Teams can have up to two incorrect letters total on their cryptogram and still be correct. The frequency of the incorrect letter is irrelevant. See the example below.
   If the cryptogram was as shown:
   KZBAOF KFXMFXYF
   SAMPLE SENTENCE

   and the students answered (underlined letters indicate mistakes)
   SAMPL<u>F</u> S<u>F</u>NT<u>F</u>NC<u>F</u>

   then it counts as four mistakes (even though the mistake was only in the letter E) and the answer DOES NOT count. However, if they put
   SAMP<u>UL</u> SENTENCE

   It is considered correct with two letter mistakes.
2. For Cryptarithms and questions that ask for the keyword/key phrase to be decoded, no "free" mistakes are allowed, and each error will result in a penalty as described in the "As you score each question" section below. These questions are denoted with the symbol ≡ on the scoring sheet as a courtesy reminder for graders.
3. When teams are asked to provide a keyword/key phrase for an aristoctat, there will be a set of boxes under the question which say "Enter the Keyword here" or "Enter the Key Phrase here". For these problems, only the answer in those boxes is to be graded (the actual cipher solution is ignored) and they are not allowed any mistakes for the full score. For each letter mistake, 100 points are to be subtracted until the score is zero.
4. Teams do NOT have to fill in the frequency table. It is simply there as an aid to them solving the cryptogram. It is NOT to be graded. It is included in the answer key as an aid to the grader.
5. When scoring the Baconian ciphers (with strange text or symbols), they can write the answer under the Baconian symbols or on the line provided. Note that you will see lots of As and Bs, but they are not graded as the answer, only what they put on the answer line.
6. As you score each question, if correct, put the number of incorrect letters (0, 1, or 2) next to the question number on the scoring page. Also, put the value for the question into the score column. If they get more than 2 letters wrong, subtract 100 points from the score until it would be zero. If a question is worth 240 points and they get 4 letters wrong, you would start with 240 points (for up to 2 letters wrong) and then subtract 100 points for each of the next two letters wrong ending up with a final score of 40 points for that question. If they had gotten 5 or more letters wrong on a 240 point question, they would receive 0 points for that question. With a 650 point question, they could get 8 letters wrong and receive 50 points (2 free letters then 6×100=600 points off). Just put the incorrect cost deduction on the score sheet and subtract it from the value for the question. Under no circumstance should the score for any question be less than zero. Note that while the timed question must have 2 or fewer letters incorrect in order to get the timing bonus, a team solving the timed question after the 10 minutes passed would be accepted as correct with 3 incorrect letters receiving 100 points for the timed question.
7. When scoring questions, if they actually did any work on the question and got some letters right, it is useful to write "LOTS" for the question on the score sheet. In this way it is really easy to break ties without having to go look at each question.
8. On a test with Special Bonus Questions (indicated by a ★), if they receive full score for the corresponding question, they will receive an additional bonus depending on the number of special bonus questions they answered. If they answered any one with no deduction points they receive 150 extra points. If they answered any two they receive 400 extra points. If they answered all three then they receive 750 extra points. A legend is at the bottom of the score sheet to make it easy to figure out the extra bonus.

9. If they correctly answered the timed question in 10-minutes or less with 2 or fewer letters incorrect, you need to compute the bonus time. Take the value for the minute from this first table below

0:xx	1,080
5:xx	480

1:xx	960
6:xx	360

2:xx	840
7:xx	240

3:xx	720
8:xx	120

4:xx	600
9:xx	0

and then add the seconds value from this table:

X:00	120
X:06	108
X:12	96
X:18	84
X:24	72
X:30	60
X:36	48
X:42	36
X:48	24
X:54	12

X:01	118
X:07	106
X:13	94
X:19	82
X:25	70
X:31	58
X:37	46
X:43	34
X:49	22
X:55	10

X:02	116
X:08	104
X:14	92
X:20	80
X:26	68
X:32	56
X:38	44
X:44	32
X:50	20
X:56	8

X:03	114
X:09	102
X:15	90
X:21	78
X:27	66
X:33	54
X:39	42
X:45	30
X:51	18
X:57	6

X:04	112
X:10	100
X:16	88
X:22	76
X:28	64
X:34	52
X:40	40
X:46	28
X:52	16
X:58	4

X:05	110
X:11	98
X:17	86
X:23	74
X:29	62
X:35	50
X:41	38
X:47	26
X:53	14
X:59	2

For example if they solved the time question at the 6:46 mark, you would add 360 (from the 6:xx entry in the first table) to 28 (from the X:46 entry in the second table) to get a bonus of 388. If they had solved it in exactly 4:00 minutes, you would add 600 and 120 to get a bonus of 720.

10. Add up all the scores and put the total on the bottom of score sheet.
11. You must break all ties. Indicate the tie breaker by adding .1 to the score of the team ahead. With multiple teams tied, you will add more. I.e. if five teams all scored 200 points, the final scores that you would enter on the score sheet would be 200.4, 200.3, 200.2, 200.1 and 200.
12. To determine how to break the tie, you need to look at the correctly answered questions in the order from the table below. If both teams answered the same (i.e. they answered the question with zero mistakes) then you go on to the next question. If one team had no mistakes and the other team had one mistake, then the team with no mistakes is ahead. For example, if one team answered question the highest value question and another team didn't, the first team will be ahead. Note that if a test is completely blank, with no evidence of work on any of the ciphers, it should be awarded a score of "P".

Tie Breaker Order	Question #
1	11
2	3
3	17
4	21
5	7
6	20
7	1
8	14
9	15
10	13
11	Timed
12	22
13	19
14	18
15	16
16	5
17	12
18	2

Tie Breaker Order	Question #
19	9
20	4
21	8
22	6
23	10

13. If there is still a tie (typically when you have teams which answered either zero, one or two questions) then you will need to look at the tie breaker questions again and count the number of correctly answered letters. The team with the most correctly matched letters is to be ahead.

Timed Question **[292 points]** Hidden in the margin of a library book on black holes: decode this Aristocrat by Henry Thoreau. What does it say? When you have solved it, raise your hand so that the time can be recorded and the solution checked.

AOZURZUE NJGCO NZGS J VZUGL VYBGS ZI GSC MYWA
DRINKING WATER WITH A MINTY MOUTH IS THE COLD

HCOIZYU YQ ITZML. DYGS VJRC VC BUMYVQYOGJDWC ZU J
VERSION OF SPICY. BOTH MAKE ME UNCOMFORTABLE IN A

NJL Z OCITCMG DBG MJU'G CUKYL.
WAY I RESPECT BUT CAN'T ENJOY.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	2	3	9	3	1		10	1	4	7	1	4	5	3	5		2	2	4	2	8	5	2		8	9
Replacement	D	U	E	B	G	Z	T	V	S	A	J	Y	C	W	R	X	F	K	H	P	N	M	L	Q	O	I

1) **[333 points]** Someone left a paper with the message below on the copy machine in the office. It's a quote by Martin Luther King Jr, encoded as a Checkerboard with a polybius key of **POSTPULMONARY**. What does it say?

AO	NC	LO	AO	AE		LC	ON	NN	NO	OO
T	H	A	T	O		N	E	K	I	D

NO	LC	ON	LR	ON		LE	ON	LC	AO	LO
I	N	E	L	E		M	E	N	T	A

LN	OR	AC	OC	NC		AE	AE	LR	AO	NC		LO	AO	OC	AE	AN		LR	OO	NO	LC	GE
R	Y	S	C	H		O	O	L	T	H		A	T	C	O	U		L	D	I	N	V

ON	LN	AO	NC	NO		AC	ON	OR	ON	LR		NO	OO	AC	AC	OC		LO	LN	ON	OO	LE
E	R	T	H	I		S	E	Y	E	L		I	D	S	S	C		A	R	E	D	M

ON
E

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2) **[226 points]** This string of symbols appears on the display of a malfunctioning calculator. It hides a quote by George Washington, encoded as a Complete Columnar. The quote has **ACADEMIC** somewhere in it. What does it say?

WCTOT UXOUI IBAXL ASLTS XCHES ETEHO MSYHR STDEK

RMOTC SENXI ARWEM X

Answer: _SCHOOL WITHOUT ACADEMIC STRESS IS LOWKEY BETTER THAN SUMMER_

3) **[542 points]** A phrase by Dalai Lama has been encoded using the Patristocrat Cipher using a K2 alphabet. You are told that the cipher contains **civilization**. What does it say?

VLJGX BGERK XBNLH EZFVL JGXBG XGHFJ HEIJS SPRHL
COUSI NSHAV INGOT HERCO USINS ISTRU THFUL LYATO

YIXKZ TZHFR PRSXB HEZZB HXFZE XGHLF PLIEJ ARBUX
PFIVE BETRA YALIN THEEN TIREH ISTOR YOFHU MANKI

BWRBW ALWZF BVXKX SXQRH XLB
NDAND MODER NCIVI LIZAT ION

Cousins having other cousins is truthfully a top five betrayal in the entire history of humankind and modern civilization.

Replacement	R	T	V	W	Z	I	N	E	X	C	U	S	A	B	L	Y	D	F	G	H	J	K	M	O	P	Q
K2	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	2	10			6	6	6	9	3	5	3	8		1		3	1	7	4	1	1	3	3	13	1	7

★(Special Bonus Question) 4) **[197 points]** Your smart fridge started flashing this cryptic message at 3 a.m. It's by Bill Gates encoded in a Baconian. What does it say?


BABAAAABAAAAAAABAAAAABAAAABBAAAABBAAAAAAAABBBABBABABBABAAABB

W E A R I N G A H O O D


ABAAAAABAABABAAABAAABAABAAABBBABBAAABBABBAAABAABBBABAAABAAAA

I E W I T H N O S H I R


BAABAABAAABAAABBABAAAABAAABAAABAAAAAAAABB

T I S W E I R D

wearing a hoodie with no shirt is weird

5) **[239 points]** Archaeologists unearth a stone tablet inscribed with a message. It's a quote by Albert Einstein encoded as an Aristocrat using a K1 alphabet. What does it say?

1111 GSFVWHFWQR ZO 1111 ZQWYB 1234321 WK VCQ
1111 MULTIPLIED BY 1111 BEING 1234321 IS THE

GUVCQGUVWTUF QISWLUFQYV XA MCWVQ FWBCV RWKHQJKWYB
MATHEMATICAL EQUIVALENT OF WHITE LIGHT DISPERSING

VCJXSBC U HJWKG VX GUEQ U JUWYZXM.
THROUGH A PRISM TO MAKE A RAINBOW.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	4	6		1	5	5	3	1	4	4	1	2		1		9	2	3	1	8	9	12	4	4	3
Replacement	F	G	H	J	K	L	M	P	Q	R	S	V	W	X	Y	Z	E	D	U	C	A	T	I	O	N	B

6) **[186 points]** You found the following carved into the bark of a hollow log. You are told that it's a quote by Mahatma Gandhi encoded as a Hill with a keyword of ENVY. What does it say?

$$\begin{pmatrix} E & N \\ V & Y \end{pmatrix} \equiv \begin{pmatrix} 4 & 13 \\ 21 & 24 \end{pmatrix}$$

W	E	H	R	T	M	U	O	Q	L	E	V	F	R	I	C	I	Z	A	G	Q	W
M	**U**	**F**	**F**	**I**	**N**	**S**	**A**	**R**	**E**	**B**	**A**	**L**	**D**	**C**	**U**	**P**	**C**	**A**	**K**	**E**	**S**

7) **[369 points]** An old robot prototype suddenly prints this encoded text. It's a quote by Giannis Antetokounmpo, encrypted using an Aristocrat using a K2 alphabet. What was the K2 Keyword used to encode it?

Keyword Answer:

BLASPHEMY

LAMVNVLT GAYH KHQJMUVLT VE J MQHHVKZQ VLNALCQLVQLNQ.
NOTICING YOUR BREATHING IS A TERRIBLE INCONVENIENCE.

QESQNVJZZG OYHVLT JL VBSAHMJLM MQEM ZVXQ MUVE.
ESPECIALLY DURING AN IMPORTANT TEST LIKE THIS.

Replacement	J	K	N	O	Q	R	T	U	V	W	X	Z	B	L	A	S	P	H	E	M	Y	C	D	F	G	I
K2	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	4	1	1		4		2	6		5	2	10	8	4	1		10		2	3	2	12		1	2	4

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8) **[192 points]** Found inside a dissected frog's stomach (plastic, don't worry): this Cryptarithm note. What do the values **6798 604** decode to?

Values to decode for solution

6	7	9	8		6	0	4
B	A	L	D		B	O	Y

Cryptarithm formula

```
  L A R R Y
  9 7 3 3 4

+ B I R D
  6 5 3 8
-----------
J O R D A N
1 0 3 8 7 2
```

LARRY+BIRD=JORDAN

	0	1	2	3	4	5	6	7	8	9
A										
B										
D										
I										
J										
L										
N										
O										
R										
Y										

9) **[201 points]** B Lou offers an observation that has been encoded as an Aristocrat using a K1 alphabet. What does it say?

ROT HR PBIT PTTQM QBDT PBIT ONEWTUM, WNH MBJ HR HTO
ONE TO FIVE FEELS LIKE FIVE NUMBERS, BUT SIX TO TEN

PTTQM QBDT PRNU ONEWTUM. B YRO'H DORS SAK, WNH BH'M
FEELS LIKE FOUR NUMBERS. I DON'T KNOW WHY, BUT IT'S

YTPBOBHTQK HUNT.
DEFINITELY TRUE.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	9		3	2			9	2	1	2		6	6	7	6	5	6	2	15	4		4		2	
Replacement	H	I	J	K	M	P	Q	T	V	X	Y	Z	S	U	N	F	L	O	W	E	R	A	B	C	D	G

★(Special Bonus Question) 10) **[88 points]** An explorer's journal mentions a “hidden truth” encoded below. It turns out to be a quote by Winston Churchill, written in a Porta Cipher using a keyword of **AGONY**. What does it say?.

A	G	O	N	Y	A	G	O	N	Y	A	G	O	N	Y	A	G	O	N	Y	A	G	O	N	Y	A	G	O	N	Y
E	L	H	A	N	R	U	K	O	G	G	X	Y	S	C	F	D	H	C	Q	E	B	U	A	Q	Q	C	H	W	Z
R	**O**	**O**	**T**	**B**	**E**	**E**	**R**	**I**	**S**	**T**	**H**	**E**	**M**	**O**	**S**	**T**	**O**	**V**	**E**	**R**	**R**	**A**	**T**	**E**	**D**	**S**	**O**	**D**	**A**

A	G	O	N	Y	A	G	O	N	Y	A	G	O	N	Y	A
B	V	U	R	X	G	Y	T	X	U	G	C	M	L	I	R
O	**F**	**A**	**L**	**L**	**T**	**I**	**M**	**E**	**I**	**T**	**S**	**T**	**R**	**U**	**E**

11) **[578 points]** Someone left this encrypted quote scribbled on the chalkboard before class began. It's a quote in Spanish, written as a Xenocrypt using a K1 alphabet. What does it say?

UZ TIESZ VXMRKIZ LTX IXSXREWZR XI WT NEYZ, XR
LA UNICA PERSONA QUE NECESITAS EN TU VIDA, ES

ZLTXUUZ LTX YXHTXRWMZ LTX WX IXSXREWZ XI UZ RTPZ.
AQUELLA QUE DEMUESTRA QUE TE NECESITA EN LA SUYA.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	Ñ	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency					4			1	6		1	4	2	1			1		7	3	8	4	1	5	15	2	11
Replacement	B	F	G	H	I	J	K	M	N	Ñ	O	Q	R	V	W	X	Y	Z	S	C	U	L	P	T	E	D	A

12) **[226 points]** Found taped under a desk in study hall: this Nihilist Substitution puzzle by Marie Curie with a keyword of **LUNCHES** and polybius key of **AUTHORISED**. What does it say?

85	24	64	76	55
P	**U**	**R**	**P**	**L**

48	57	82	24	67
E	**G**	**L**	**U**	**E**

55	27	46	55	76
S	**T**	**I**	**C**	**K**

35	54	53	38	55
S	**A**	**R**	**E**	**B**

47	54	25	67	53
E	**T**	**T**	**E**	**R**

27	38	34	84	64
T	**H**	**A**	**N**	**W**

57	54	27	48	38
H	**I**	**T**	**E**	**O**

84	36	66
N	**E**	**S**

13) **[292 points]** A message flashes briefly on a malfunctioning computer screen. It turns out to be a quote by Ada Lovelace written in an Aristocrat. What does it say?

QBNEAPX N XTTV GTPX TP KQB ENVAT AG UALB GBBAPX N
HEARING A GOOD SONG ON THE RADIO IS LIKE SEEING A

JAUV NPAONU AP AKG PNKYENU QNIAKNK, IYK QBNEAPX AK
WILD ANIMAL IN ITS NATURAL HABITAT, BUT HEARING IT

AP N ZUNCUAGK AG UALB GBBAPX AK AP N RTT.
IN A PLAYLIST IS LIKE SEEING IT IN A ZOO.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	19	9	1		4		7		2	1	9	2		14	1	11	4	1		7	7	3		6	2	1
Replacement	I	E	Y	C	R	V	S	X	B	W	T	K	F	A	M	N	H	Z	Q	O	L	D	J	G	U	P

14) **[332 points]** You receive an anonymous text message containing a coded passage. It's said to be a quote by Curt Cignetti, encrypted with a Complete Columnar. The quote has **OPPOSITE** somewhere in it. What does it say?

NASET LEMAM OHUBH EREEE DAASR EOOEA RLEEP NLNEE

DNPTB DTNGB THEEE EDYIW RURI

Answer:
RED AND GREEN MAY BE OPPOSITE ON THE WHEEL, BUT RED AND BLUE ARE THE REAL ENEMIES

—

15) **[313 points]** This code is found written in chalk on the sidewalk outside the science building. It's a quote by Charles Darwin, written in a Baconian Cipher. You are told that the deciphered text ends with **nest**. What does it say?

```
AN ARK AGAIN CARRY AN EGG CAMEL AN ELK ACTED ALONE
AAAAB  ABAAA BAAAA AAABB  BAAAB AAABB  ABBAB ABBAA

  B      I     R     D      S     D      O     N


CAUSE ARDOR ANGIE CAUSE ANGRY ALLAN ASKED CENTS VISOR
BAABA AABBA AABAA BAABA AABAA ABBAA ABBAB BAABB AABBA

  T     G     E     T     E     N     O     U     G


BACKS VERGE SAMMY UNDER RANCH IS ANY LEMON JULIO
AABBB AAABA BAAAA AABAA AAABB ABAAA  BAABA AABAB

  H     C     R     E     D     I      T     F


RODEO HAVEN MASKS VOCAL PETEY MASKS FUZZY HAVIN FINDS
ABBAB BAAAA AABBB ABBAB BABAA AABBB AAAAA BAAAA AAABB

  O     R     H     O     W     H     A     R     D


RHYME LEROY ELIZA GRAIL LEFT A RODEO FRAUD SEALS
ABAAA BAABA ABAAA BAAAB BAABA  ABBAB AAAAB BAABB

  I     T     I     S     T      O     B     U


MOURN ROUTE FEEDS FEVER JOHAN METER KNEEL LABOR
ABAAA ABABA AAABB AAAAA ABBAA AABAA BAAAB BAABA

  I     L     D     A     N     E     S     T
```

Birds don't get enough credit for how hard it is to build a nest

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16) **[246 points]** A coded message appears after you shine a UV light on a classroom poster. It turns out to be a quote by Henry Ford, encrypted using a Nihilist Substitution with a keyword of **MANUEL** and polybius key of **PROMULGATED**. What does it say?

38	58	68	28	68
T	**H**	**E**	**O**	**N**

42	68	55	66	46
L	**Y**	**B**	**A**	**D**

36	44	26	47	66
P	**A**	**R**	**T**	**A**

47	38	36	38	57
B	**O**	**U**	**T**	**F**

66	36	46	62	57
A	**L**	**L**	**I**	**N**

45	84	60	47	46
G	**I**	**S**	**G**	**E**

38	47	84	58	47
T	**T**	**I**	**N**	**G**

56	29	35	67	28
H	**U**	**R**	**T**	**O**

49	56	39	35	95
T	**H**	**E**	**R**	**W**

56	70	46	55	47
I	**S**	**E**	**I**	**T**

88	27	50	44	35
S	**R**	**E**	**A**	**L**

44	97	49	40	64
L	**Y**	**F**	**U**	**N**

17) **[495 points]** You receive this cryptic message from the future through an online time capsule. It's a quote by Sun Tzu, written in a Baconian. What does it say?

```
rdoayudagyhdaroughydrougharydouaghryougdhraydoayud
ABABBABBABABBAAAAABBAAAAABABBAABAAABAAABAABBBABBAB

  M    O    N    D    A    Y    I    S    H    O

gahyrougdahyrougdahrouydgahroydugayhdaroughydrough
ABABAAAABBABAAAABBAAAABBABAAABBAABBABBAAAAABBAAAA

  L    D    I    N    G    S    U    N    D    A

arydoughraoughrougyhrdouaghyrdaoughrydoauygdahryda
BABBAAAAABAAAAAAAABAABAABAABABBAAAAABBABABABBAABBB

  Y    B    A    C    K    F    R    O    M    H

ouyghdrougahryodughraouydgharouydaghydrayoudghraou
AABAABAAAABAABABAAAABAABBAABAAABBBAABBABBAABAAABAA

  E    R    T    R    U    E    P    O    T    E

gydhraouyghdroughrougahyr
ABBAABAABAABAAAAAAAAABABA

  N    T    I    A    L
```

Monday is holding Sunday back from her true potential

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18) **[253 points]** Solve this quote that has been encoded using the Cryptarithm Cipher. What do the values `34516 30862` decode to?

Values to decode for solution

3	4	5	1	6
S	O	L	A	R

3	0	8	6	2
S	H	I	R	T

Cryptarithm formula

```
  A R C T I C
  1 6 7 2 8 7

    F R O S T
- 9 6 4 3 2
-----------
  C H I L L
  7 0 8 5 5   ARCTIC-FROST=CHILL
```

	0	1	2	3	4	5	6	7	8	9
A										
C										
F										
H										
I										
L										
O										
R										
S										
T										

19) **[254 points]** Srinivasa Ramanujan offers us some advice that has been encoded as an Aristocrat. What does it say?

OUEOVKEQQT, YORFQGCB XFCAT KUIZ F LFLT GY ARKGCGORQT
TRUTHFULLY, STEALING CANDY FROM A BABY IS DEFINITELY

CIO FY RFYT FY GO YIECAY. OUEYO ZR. G YHRFJ KUIZ
NOT AS EASY AS IT SOUNDS. TRUST ME. I SPEAK FROM

RDHRUGRCXR.
EXPERIENCE.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	3	1	6	1	4	8	7	2	4	1	4	2			8		4	10		5	5	1		2	9	3
Replacement	D	G	N	X	U	A	I	P	O	K	F	B	Q	W	T	J	L	E	V	Y	R	H	Z	C	S	M

★(Special Bonus Question) 20) **[345 points]** Gordon Ramsay has been heard to say the following phrase that has been encoded using the Checkerboard Cipher with a polybius key of **SUBCONTIGUOUS**. What does it say?

NA	ME	NA	MD	ME	NE	ES	MS	NH	AH
P	**O**	**P**	**C**	**O**	**R**	**N**	**S**	**M**	**E**

NS	NS	MS	MS	EA	ED	ES	EA	AA	EA
L	**L**	**S**	**S**	**I**	**G**	**N**	**I**	**F**	**I**

MD	EE	ES	EH	NS	SD	MA	AH	EH	EH	AH	NE	EH	AD	EE	ES	AD	ME	SH	EA
C	**A**	**N**	**T**	**L**	**Y**	**B**	**E**	**T**	**T**	**E**	**R**	**T**	**H**	**A**	**N**	**H**	**O**	**W**	**I**

EH	EH	EE	MS	EH	AH	MS	AH	MS	NA	AH	MD	EA	EE	NS	NS	SD	EE	EH	EH
T	**T**	**A**	**S**	**T**	**E**	**S**	**E**	**S**	**P**	**E**	**C**	**I**	**A**	**L**	**L**	**Y**	**A**	**T**	**T**

AD	AH	NH	ME	SS	EA	AH	MS
H	**E**	**M**	**O**	**V**	**I**	**E**	**S**

21) **[478 points]** Someone passed you a piece of paper with this Aristocrat encoded quote by Mother Teresa using a K3 alphabet. What was the K3 Keyword used to encode it?

Keyword Answer:

DOMINATES

FQG BZEJGY GSPQF ATGHB'F YGKGSDG GBTZPQ KYGASF LTY
THE NUMBER EIGHT DOESN'T RECEIVE ENOUGH CREDIT FOR

KYGCFSBP GDGYI TFQGY BZEJGY TY ASPSF TB C HKTYGJTCYA.
CREATING EVERY OTHER NUMBER OR DIGIT ON A SCOREBOARD.

K3	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	4	6	3	2	2	7	15	2	1	3	4	1				4	4		6	8					11	3
Replacement	D	N	A	V	M	T	E	S	Y	B	C	F	X	Z	W	G	H	J	I	O	K	L	P	Q	R	U

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22) **[278 points]** While scrolling through an old floppy disk labeled "TOP SECRET", you discovered this quote by Terry Davis hidden as an Aristocrat. What does it say?

TMZIQ RZHAME NXVB ZI QKY HNXVV ZV JIM JS BWM YJVB
BEING PICKED LAST IN GYM CLASS IS ONE OF THE MOST

MYTXPPXVVZIQ BWZIQV X NZFZIQ JPQXIZVY HXI MIELPM.
EMBARRASSING THINGS A LIVING ORGANISM CAN ENDURE.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	4			2	1		3	9	4	1	1	7	3		4	6	1	1	2		9	2	7	4	9
Replacement	K	T	J	Q	D	V	Z	C	N	O	Y	U	E	L	X	R	G	P	F	B	W	S	H	A	M	I

BullSO, University of South Florida - Repaired practice edition - Answer key

Division C | Codebusters | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

repair-case. (2 points) Authored written substitute for unavailable source material: Which operation preserves the exact letter counts while changing order?

A. Monoalphabetic letter replacement

B. Deleting spaces and letters

C. A numerical hash

D. Transposition

Correct: D